

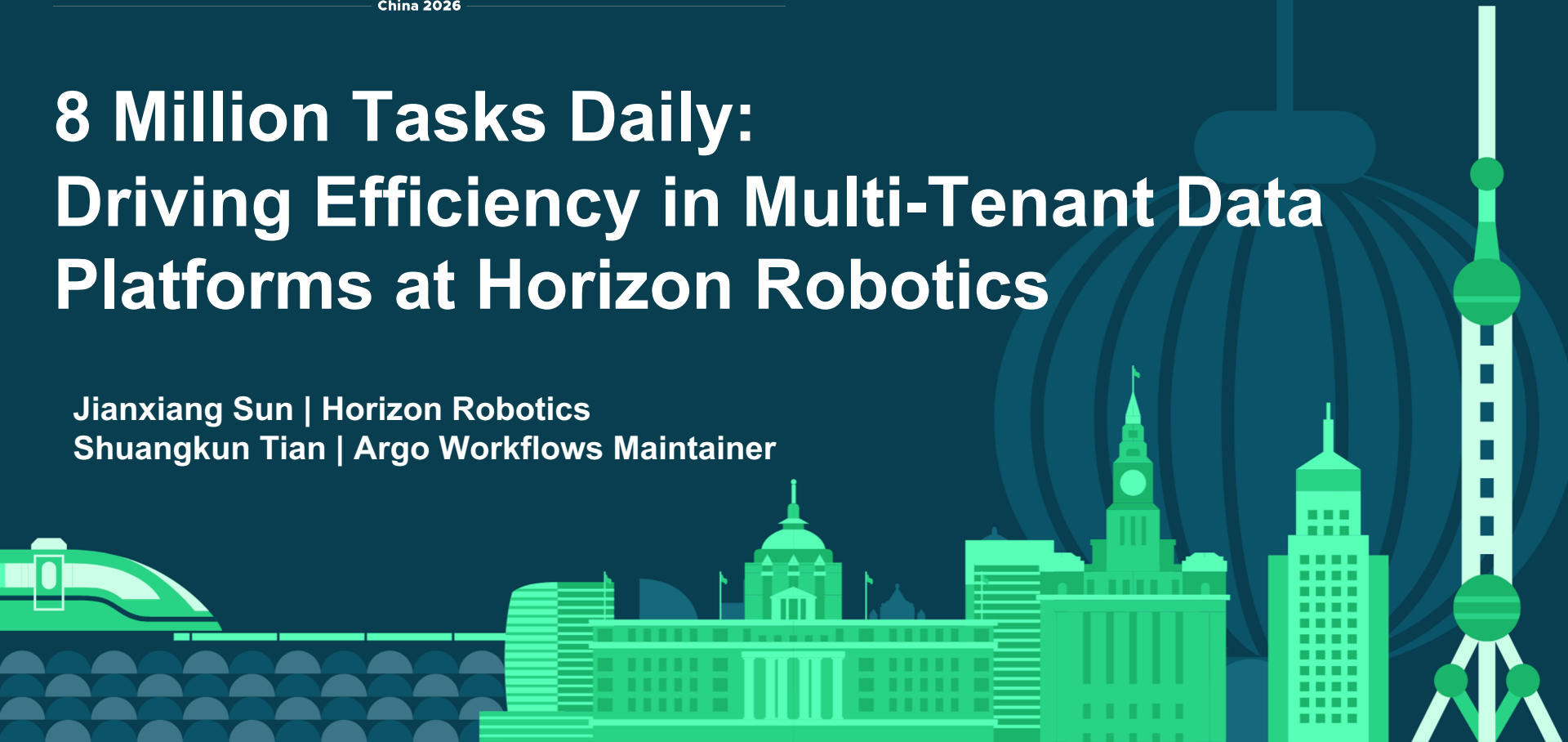


#KubeCon #CloudNativeCon
#OpenInfraSummit #PyTorchCon

China 2026

8 Million Tasks Daily: Driving Efficiency in Multi-Tenant Data Platforms at Horizon Robotics

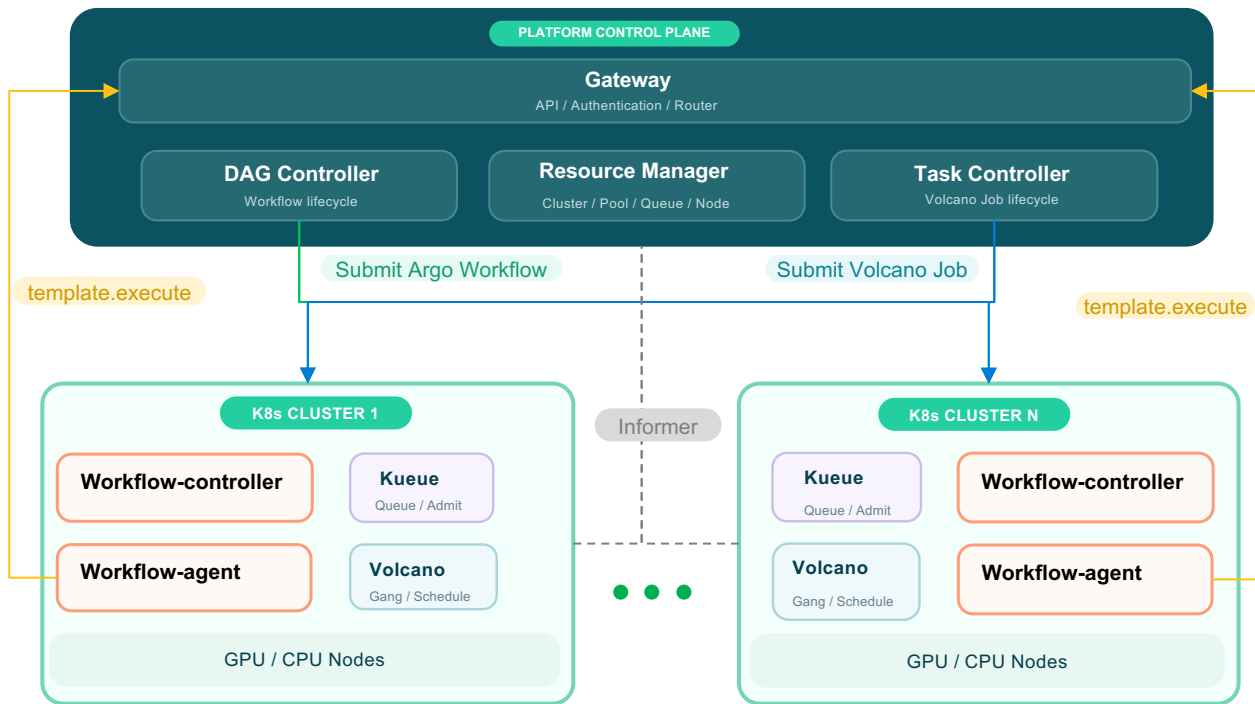
Jianxiang Sun | Horizon Robotics
Shuangkun Tian | Argo Workflows Maintainer



Platform Architecture



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8M tasks daily

1min task duration

30K workflows per cluster

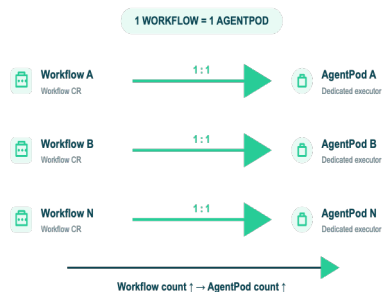
100K pods per cluster

Challenge



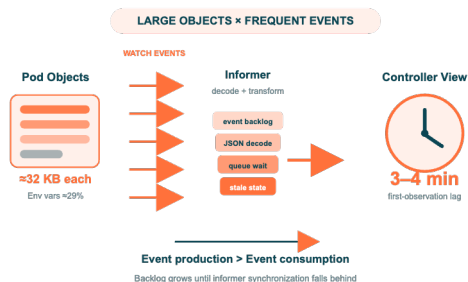
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01 Infra Tax



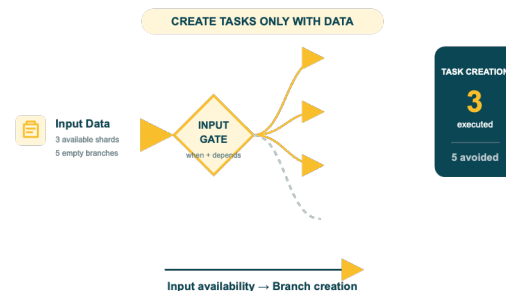
More Workflows, More AgentPods,
More Machines

02 Informer Choke



Large Objects, Short Task

03 Wasted Compute



Fixed Concurrency, Less Data Slices

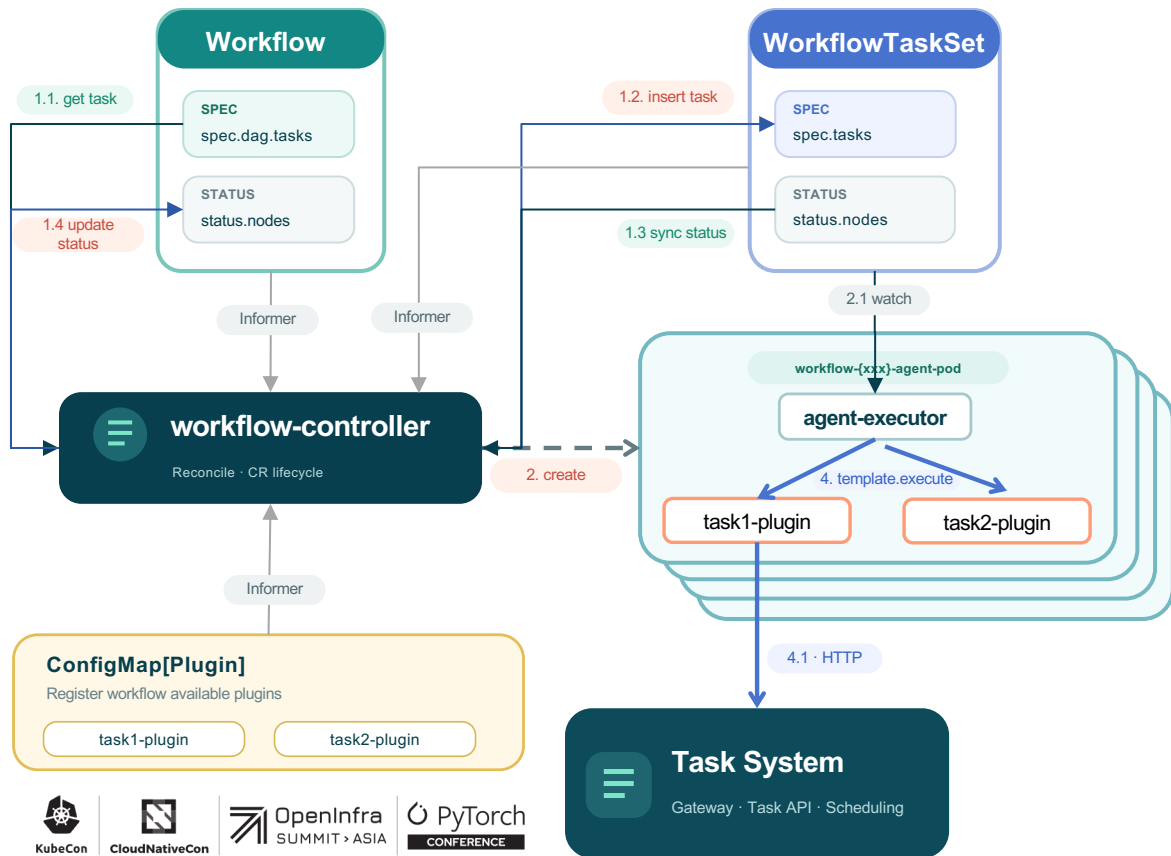


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01 Infra Tax



Argo Executor Plugin Architecture



Benefits

- Dynamic plugin registration
- Fault isolation
- WorkflowTaskSet as channel

Disadvantages

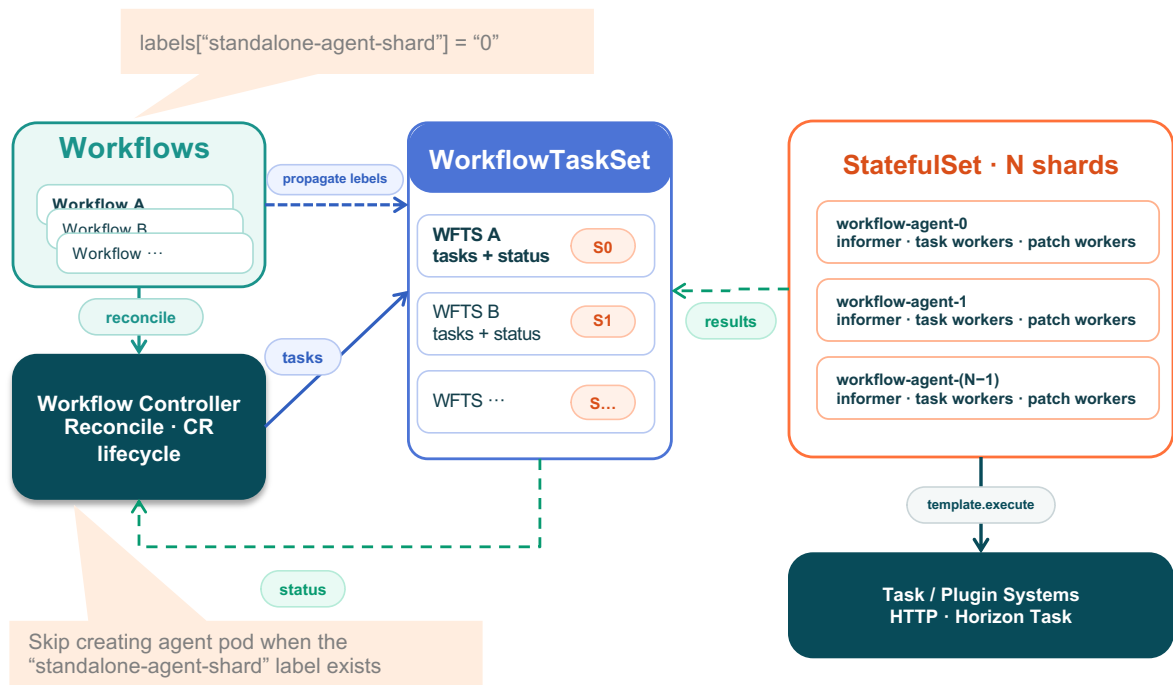
- AgentPod count scales linearly with workflows
- Too many watch connections to apiserver

Agentless Architecture

CASE 01 / INFRA TAX



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Share

10k+ Workflows share a small, long-running workflow-agent instance.

Shard

Use $\text{hash}(\text{workflow}) \% N$ to assign stable ownership to exactly one agent shard.

Bound

A fixed agent pool caps infrastructure usage, while supporting the cluster's planned workflow capacity.

Agentless Benefits



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CASE 01 / INFRA TAX

IMMEDIATE IMPACT

10K+

Workflows / instance

5

Instances per cluster

1,000+

Nodes eliminated

Operational dependency removed

No longer depends on elastic-node availability or emergency vendor provisioning.

FROM SAVINGS TO PLATFORM CAPABILITY

CAPABILITY UNLOCKED

Standardized and Predictable Cluster Capacity

01

A consistent capacity model makes available headroom measurable, comparable, and plannable.

Capacity-aware Workflow placement

02

Workflows are dynamically routed to clusters with available capacity.

Global scheduling foundation

03

Balance utilization across clusters and improve platform resilience.

Infrastructure savings are the outcome. Schedulable capacity is the foundation.

Use Agentless only inside a clear operating envelope

Three gates — all must be true

01

Stateless and retry-safe

No long-lived in-agent state; a task can be safely retried.

02

Short-lived or I/O-bound

HTTP, polling and plugin calls; high volume of short workflows.

03

Fits fixed shard capacity

Demand stays within the cluster workflow limit and fixed N agents.

ALL
YESANY
NO

Decision

USE AGENTLESS

Shared agents remove per-workflow pods and keep the footprint predictable.

KEEP ISOLATED

Long-running state, strict isolation, or demand beyond the shard budget.

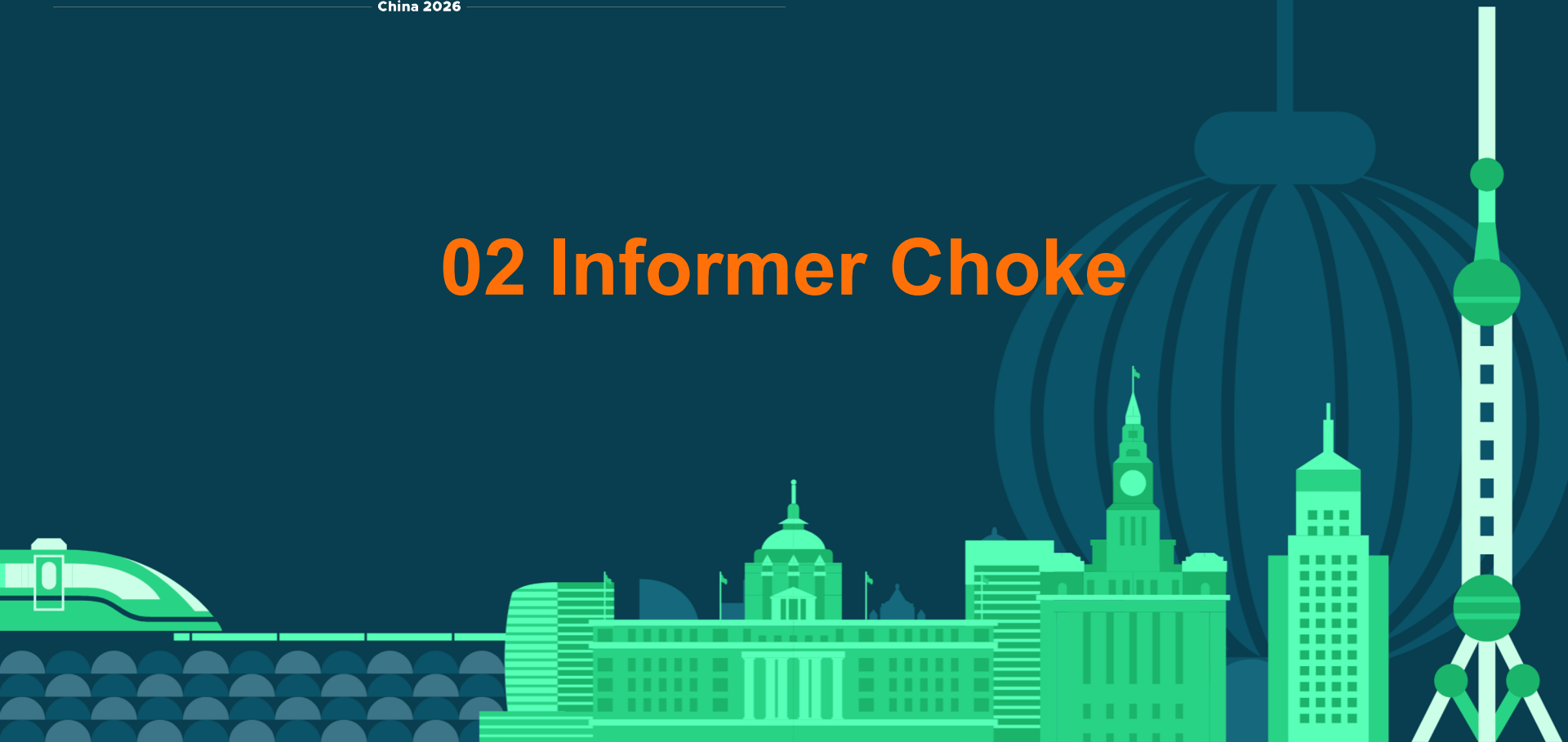
Production guardrails

- Watch workflows and tasks per shard
- Bound task/patch workers and Kubernetes API QPS
- Plan shard-count changes; running workflows do not rebalance



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02 Informer Choke



Initial Investigation

CASE 02 / INFORMER CHOKE



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Diagnosing Informer Latency Beyond Controller WorkQueues



WorkQueue

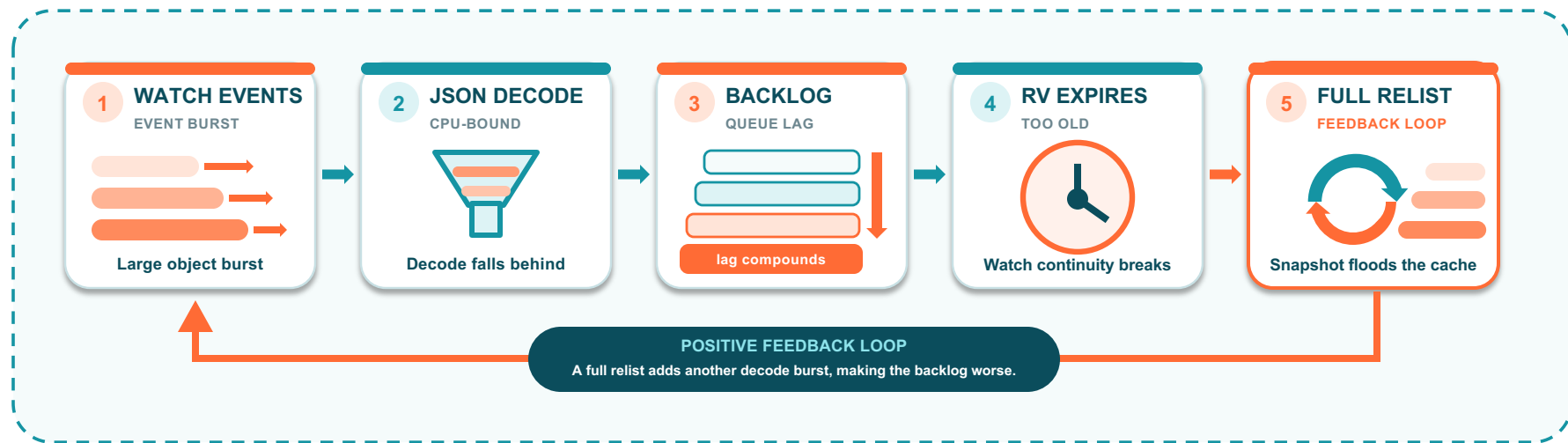
Normal

First Observation

Minutes

The bottleneck is in Informer ingestion, not reconciliation.

Events Arrive Faster Than the Informer Can Process Them



Short Tasks × Large JSON Objects × Frequent State Changes → Informer Choke



INFORMER FALLS BEHIND

Speed Up Deserialization



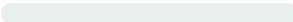
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CASE 02 / INFORMER CHOKE

01 Decode Faster

50–80%

smaller payloads

JSON 

PB 



2–6× faster encode / decode

Prefer Protobuf
JSON fallback for CRDs

02 Move Fewer Bytes

VJ object · ~32 KB

Other · 70%

ENV
30%

~10 KB

environment variables

Inline ENV



ConfigMap

Externalize ENV
Smaller VJ + Pod objects

03 Split Event Streams

Independent List/Watch streams

shard=A

shard=B

Informer A

Informer B

List / Watch

List / Watch

PARALLEL JSON DECODE

Argo: InstanceID · Volcano: custom shard label

LESS DECODE CPU



FEWER BYTES



PARALLEL WATCHES



LOWER INFORMER LAG

Benefits

CASE 02 / INFORMER CHOKE



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1s

Typical informer latency

-50%

Watch event size

-37%

Data mining batch time



Lower informer lag · No ReList observed

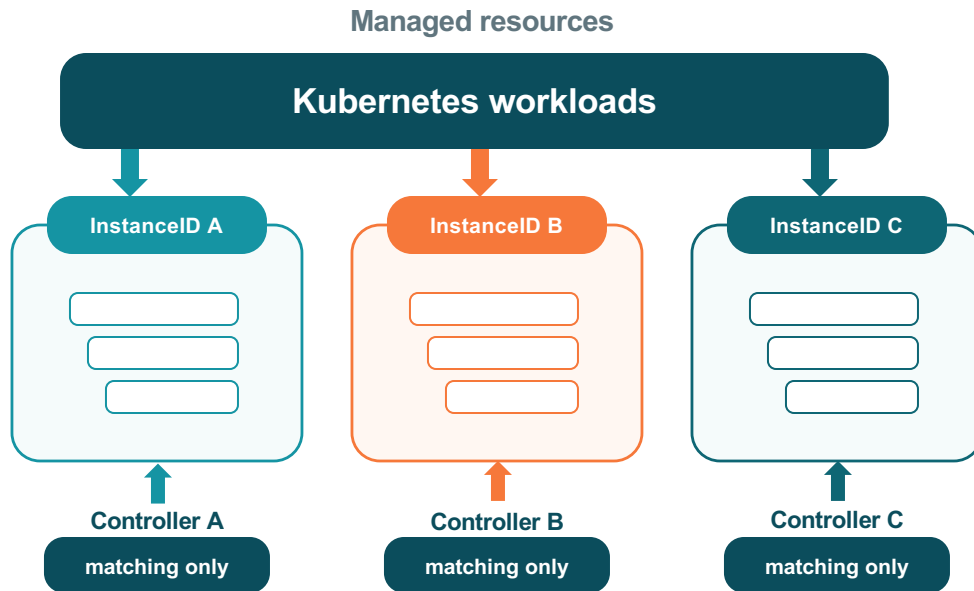
Argo InstanceID Is a Natural Isolation Boundary

ARGO INSTANCEID

Labels managed resources with
controller ownership

Scopes each controller to matching
resources

A built-in isolation boundary —
and a natural shard key





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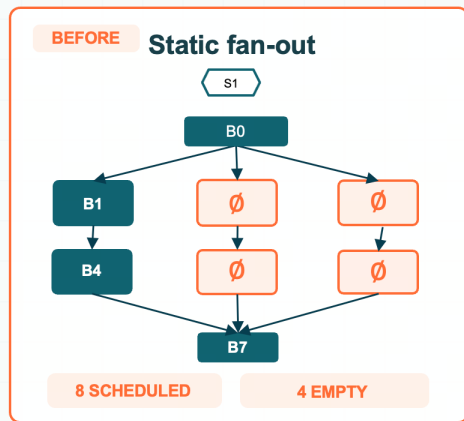
03 Wasted Compute



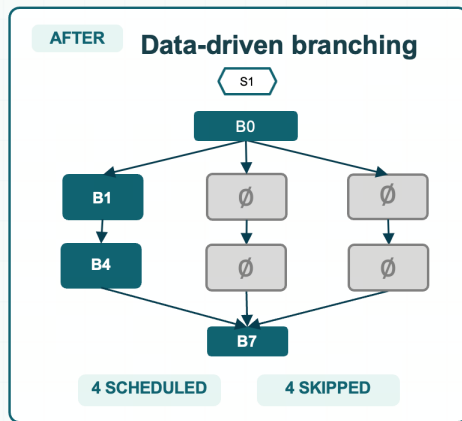
Eliminating Empty Tasks with Argo Conditional Execution

8 launched → 4 launched

Data determines branches before task creation



inspect
inputs



Skipped branches never become tasks — no scheduling, startup, or empty execution.

-57.5%

fewer empty tasks in production

**3 predefined branches
Only 1 input shard**

Before: all 8 start
After: only 4 execute

Argo Conditional Execution: outputs, when, and depends

outputs

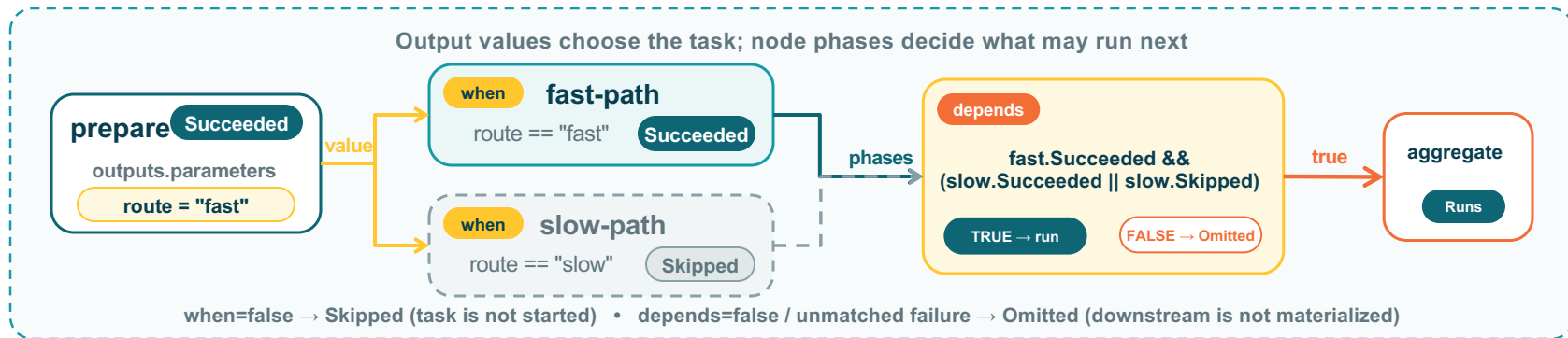
Expose results from upstream nodes

when

Decide whether the current node runs

depends

Define downstream behavior for Skipped / Failed



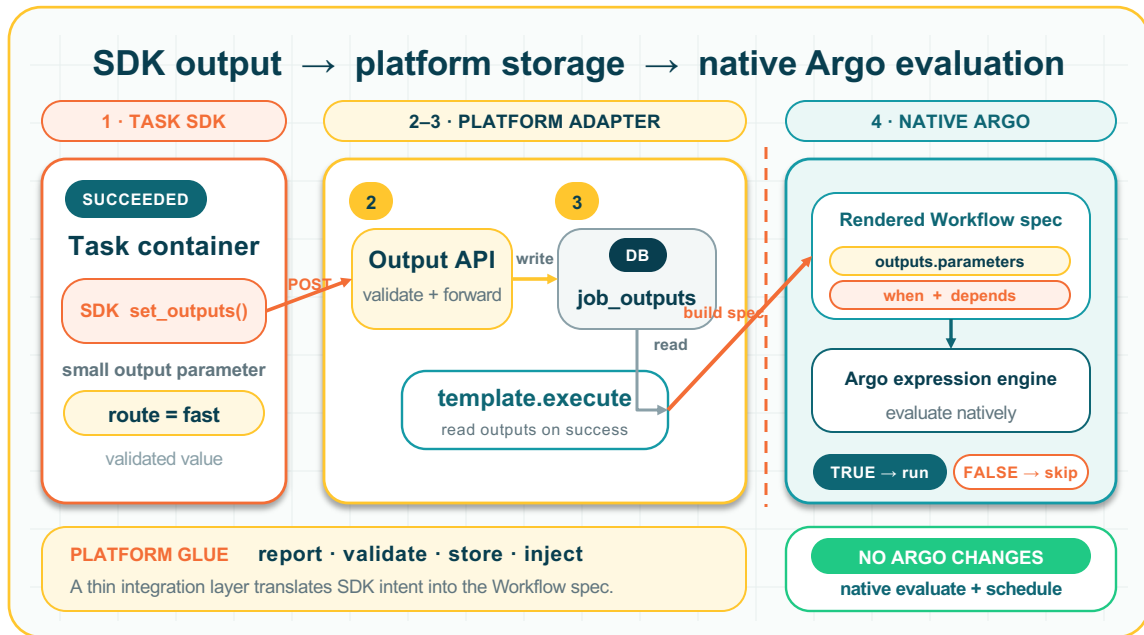
Argo Workflows Condition



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CASE 03 / WASTED COMPUTE

Platform Integration for Conditional DAG Execution



Integration Boundary

- SDK reports small output parameters
- Platform stores and returns Outputs
- Workflow builds when / depends
- Argo evaluates native expressions

No changes to Argo itself

KEY TAKEAWAYS

Three Lessons at Scale

- 01 | Focus on the infrastructure costs that grow fastest with workload scale.
- 02 | Treat cluster communication as a first-class scalability bottleneck.
- 03 | Optimize around real workload needs—not just lower-level performance.

Q&A

Jianxiang Sun · Horizon Robotics
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